Daniel Rushakoff

dzrushakoff@cpp.edu | 310-292-3714 | https://portfolium.com/DanielRushakoff | https://www.linkedin.com/in/daniel-rushakoff | Redondo Beach, CA | US Citizen | Security Clearance Eligible

EDUCATION

Cal Poly Pomona - Bachelor of Science in Electrical Engineering **December 2025**

- **GPA:** 3.66 | Dean's and President's Honor Lists
- **Affiliations:** IEEE Student Member

Santa Monica College - Associate of Arts in General Science **August 2023**

- Dean's Honor List | Graduated with Honors

SKILLS

Programming: C, C++, Python, PIC Assembly (MPASM), MATLAB, Verilog
Hardware & Embedded: Arduino, PIC, Raspberry Pi, FPGA, UART, I^2C, SPI, PWM, ADC/DAC
Software: Simulink, LTspice, PSpice, Cadence OrCAD, LabVIEW, MPLAB X IDE, KiCad, Altium

RELEVANT COURSEWORK

Systems, Controls & Signal Processing: Control Systems Engineering, Digital Signal Processing, Signals and Systems, Communication Systems, Probability and Random Processes, Biomedical Signals and Instrumentation.

RF, Power & Analog Electronics: Electromagnetic Fields, Power Electronics, Power Engineering, Analog Microelectronics, Microelectronic Devices and Circuits, Circuit Analysis I & II.

Digital Logic & Embedded Computing: VLSI Circuit Design, Microcontrollers, Digital Logic Design, C for Engineers.

PROJECTS

Semi-Autonomous Environmental Sensing & Terrain Mapping UAV [▸VIEW] Jan. 2025-Dec. 2025

- Engineered a custom mixed-signal PCB integrating MQ-2, MiCS-5524, and BME280 sensors with a high-precision ADC (MCP3008) to establish a robust I2C data pipeline to the Raspberry Pi 5 host.
- Developed dual-camera optical terrain navigation for real-time path planning, architecting a hardware migration roadmap for future YOLO-based computer vision integration via NVIDIA Jetson.
- Fabricated ruggedized perfboard interface using point-to-point routing for flight.

Wireless Arduino-Based ECG Heartbeat Monitoring Healthcare System [▸VIEW] September 2025

- Designed an analog dual-sensor biomedical circuit utilizing AD8232 and MAX30102 to acquire real-time ECG and SpO2 data.
- Implemented Digital Signal Processing (DSP) algorithms in C++ to filter 60Hz noise and motion artifacts, enabling clinical-grade HRV and arrhythmia detection.

Wireless PID Motor Control & Telemetry System **September 2025**

- Architected a wireless closed-loop motor control system utilizing two Arduino Uno R4 WiFi MCUs and NRF24L01+ transceivers to actuate a 25,000 RPM DC motor via an L293D H-Bridge driver.
- Developed a discrete PID control algorithm (50 ms rate) with integral anti-windup to process 8.3 kHz optical encoder pulses, engineering a LabVIEW GUI to command setpoints and monitor RF telemetry from I2C/ADC sensors (TC74A0, DHT11, LDR).

Wireless LDR Intensity Control System **[▸VIEW] August 2025**

- Designed a closed-loop SPI system utilizing Arduinos and nRF24L01+ transceivers (stabilized by 3.3V LDO regulators and decoupling capacitors) to wirelessly transmit LDR sensor telemetry.
- Developed a 9600-baud LabVIEW GUI to dynamically map 1-byte telemetry to discrete LED dimmer levels, integrating I2C 16x02 LCDs for local status monitoring.

PIC18F Environmental & PWM Fan Control System **[▸VIEW] December 2024**

- Engineered a PIC18F4620 embedded system in C with bare-metal firmware drivers (I2C, SPI, ADC, UART, NEC IR) for closed-loop DS1621 thermal PWM fan control, timer-based RPM tachometer tracking, hardware interrupts, DS3231 RTC scheduling, and live ST7735 TFT GUI updates.